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particular date, it would be much easier to find an answer to "When was Britney born".

As of now, the Semantic Web is only a vision, with its proponents and detractors. There are various technical reasons for this, which we cannot go into within the scope of this article. We'll just have to wait and watch to see if it takes off.

Emergent Semantics

Emergent semantics is described as "a self-organising alternative to the Semantic Web that does not require any recoding of the data currently available online. Based on successful experiments with communities of robots, emergent-semantic technology is built on the principles of human learning." This is being worked on by Sony Computer Science Laboratory. Could this 'emergent semantics' be a viable alternative to the Semantic Web?

In November 2004, an article by Junko Yoshida and R Colin Johnson described emergent semantics as extracting the meaning of Web documents from the manner in which people use them. The scheme would harness the human communication and social interaction among peer-to-peer file sharers, database searchers and content creators to append the semantic dimension to the Web automatically, instead of depending on the owner of each piece of data to tag it.

Sony argues that this latter method—of the owner of each Web page, for example, tagging the page with its meaning—is similar to attempting AI by writing 'if-then' statements about everything in the world.

So how would emergent semantics help automatically tag documents? A basic explanation is that the meaning of a document is taken from the browsing paths of all the people that browse that document. A browsing path, of course, is the path that you take while following document links—from Classical.com to, say, Beethoven.com, to Mp3.com. Since you visited Classical.com before and Mp3.com after visiting Beethoven.com, there is an indication that

Beethoven.com has something to do with 'classical' and with MP3s. Now take the behaviour of all the users who've visited Beethoven.com and there you have it—the page is about Beethoven, Beethoven is a person, the page has classical music on it, and so on!

Latent Semantic Indexing

Talking about 'semanticising' the Web, another important technique is Latent Semantic Indexing (LSI). With a regular keyword search, a document, looked at from the search engine's point of view either contains a keyword or doesn't. There's no middle ground. And each document stands alone—there's no interdependence between documents. The Web is not viewed for the collection of documents that it is: it is viewed as a lot of individual documents taken separately.

In LSI, the regular recording of what words a document contains is done first. The important addition is that it examines the document collection as a whole to look for other documents that may contain the same words in a certain document. What does this do? Essentially, if two documents have a lot of words in common, they are 'semantically close'. ('Semantics' means 'meaning'.) And if two documents don't have many words in common, they are semantically distant.

Now when you perform a search on a database that has been indexed by the LSI method, the search engine looks for documents that semantically match the keywords. For example, in the semantic system, we're talking about, 'crocodile' and 'alligator' are pretty close, so a search on 'crocodile' would also bring up pages that contain only 'alligator' with no mention of 'crocodile'.

If search engines were to use LSI, they would be more powerful. Think about the fact that a search engine looks for pages that contain all your keywords. If you were to enter twenty keywords into a regular search engine, you'd get very few results—but LSI shows the reverse behaviour. If you enter more search terms into a search engine that has LSI'd the Web, it's likely to find more, not less, documents of relevance—for the simple reason that it would bring up closely related documents for each keyword.

You could then filter the results according to relevance, which would provide feedback to the engine about what you think best matches your query. This, combined with personalisation (see the 'Personalisation' section in this article for more), would lead to your results getting much better over time.

LSI could also help archivists—those who categorise documents into classes based on their contents. If there were an LSI system in place, a document would already have attributes assigned to it, in the sense of "this document is about such-and-such a topic." The archivist would only have to add to or subtract from the list of attributes, instead of having to make up a list from scratch.

In the real world, it was reported on 5th February 2005 on Seobook.com, "Many people have been noticing a wide shuffle in search relevancy scores recently. Some of those well in the

Search Engine Optimisation

Search engine optimisation (SEO) is the process of effectively modifying a Web site so it shows up high in the results page by a search engine. Basically, an SEO'd site is more spider-friendly than a non-SEO'd one.

What happens when a spider visits a Web page? It looks at the title tag, the Meta tag, the 'alt' tag in images on the site, and so on. All these need to be filled in, and filled in well. 'Well' here means that they should be accurate reflections of what the site is about, and that they should be dense enough for the spider to get sufficient information from them.

There are, of course, several unscrupulous ways of boosting search engine rankings. For example, a site may put in a lot of hidden text that will drive users to the page—a

pornographic site may include hidden text about, say, "Windows", so that the page shows up along with 'Windows' results when you're looking for info on Windows. Similarly, one may repeat keywords hundreds of times on a page, so that the site ranks high for that keyword.

Take cloaking, for example. A popular definition says cloaking is "the process by which your site can display different pages under different circumstances. It is primarily used to show an optimised page to the search engines and a different page to humans." Most search engines penalise a site if and when they discover it is using cloaking.

SEO is big business, especially because of the exponential growth of the number of pages on the Web.